

DATA-EFFICIENT BREAST CELL NUCLEUS SEGMENTATION WITH EDGE PRIORS

Varun Kumar Dasoju¹, Qingsu Cheng², Zeyun Yu¹

University of Wisconsin–Milwaukee, Milwaukee, WI, USA

¹Department of Computer Science; ²Department of Biomedical Engineering
vdasoju@uwm.edu, chengq@uwm.edu, yuz@uwm.edu

ABSTRACT

Hand-crafted edge channels are often presented as data-efficient priors for nuclear segmentation, yet their gain over matched baselines and their behavior under domain shift are rarely isolated. We audit an EfficientNet-B7/UNet++ mammary-nucleus pipeline and conduct 69 matched runs on the official BBBC039 split. The Gabor-projection-SCSE bundle improves Dice over vanilla B7 by 0.250 points (96.85% vs. 96.60%; 95% CI, 0.197–0.299), but is indistinguishable from matched UNet++/Xception/SCSE (96.87%; difference –0.014, CI –0.047–0.018). A corrected source-group audit of 855 private UWM MCF10A planes reverses the earlier superiority claim: the full pipeline obtains 93.12% Dice versus 93.49% for vanilla B7. On 50 TNBC histology images, fluorescence-to-histology transfer is 8.07%; a single post-hoc polarity stress test raises it to 51.30% without training. Thus edge priors provide at most a small within-screen gain, while protocol integrity, architecture and acquisition domain dominate conclusions.

Index Terms— nucleus segmentation, Gabor filters, reproducibility, domain shift, microscopy

1. INTRODUCTION

Accurate nuclear masks support cell counting and quantitative morphology, but annotations are expensive and imaging distributions vary sharply. U-Net [1], UNet++ [2] and pretrained encoders make strong baselines; modern generalist models further raise the comparison standard [3, 4]. Consequently, a complex pipeline should be judged against equally trained controls, not historical scores obtained under different splits, budgets or metrics.

This work revisits a project that progressed from 2-D mammary epithelial nuclei to exploratory 3-D reconstruction. Its 2-D method combines a Gabor response, learned input projection, EfficientNet-B7, UNet++ and spatial/channel squeeze-and-excitation (SCSE). Earlier review identified unresolved empty-image scoring, split independence, boundary claims, unequal baselines, Gabor sensitivity, sampling details, statistical uncertainty and absent external validation. We address those issues on public data while restricting the task

to *semantic nuclear foreground segmentation*; it is neither breast-cancer diagnosis nor clinical validation.

Our contributions are: (1) a visible reconciliation of the private UWM result, including leakage, mask-encoding and arithmetic defects in the rejected analysis; (2) a leakage-checked, native-grid, multi-seed public evaluation with explicit empty-mask rules and boundary metrics; (3) matched architecture, component and Gabor/sampling controls; and (4) a complete TNBC transfer stress test showing that polarity mismatch, rather than omitted grayscale conversion, explains much of the failure. All attempted configurations and negative results are retained.

2. METHOD

2.1. Edge-enhanced model

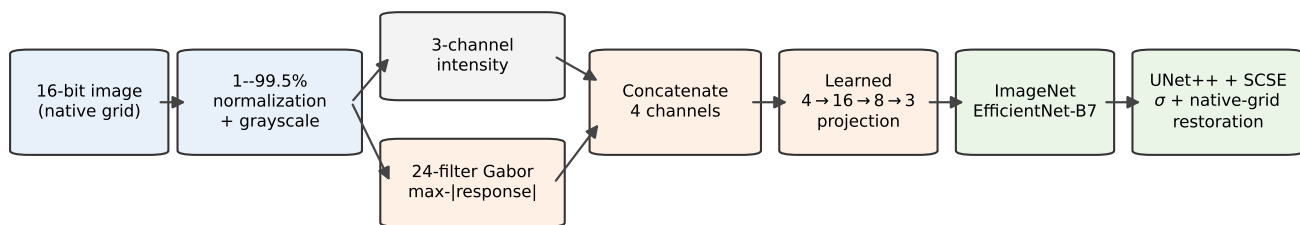
Let x be percentile-rescaled grayscale intensity. The edge channel is the maximum absolute response of 24 filters (eight orientations and frequencies $f \in \{0.05, 0.15, 0.25\}$ cycles/pixel):

$$e(x) = \mathcal{N} \left(\max_{\theta, f} \left| x * \frac{k_{\theta, f}}{\|k_{\theta, f}\|_1 + \epsilon} \right| \right), \quad (1)$$

where k has size 31, $\sigma = 5$, aspect ratio 0.5 and zero phase, and $\mathcal{N}$ maps to $[0, 1]$. L1 normalization replaces unstable signed-sum normalization. A learned $4 \rightarrow 16 \rightarrow 8 \rightarrow 3$ projection feeds ImageNet-pretrained EfficientNet-B7/UNet++/SCSE. Controls remove the edge and SCSE, replace Gabor with Sobel, or use identity-preserving residual fusion $x_{out} = x_{RGB} + \tanh(a)P([x_{RGB}, e])$ with $a_0 = 0$.

All models minimize $0.5\mathcal{L}_{Dice} + 0.5\mathcal{L}_{BCE}$. Boundary variants add $0.1\mathcal{L}_{Dice}(B(p), B(y))$, where $B(u) = \text{maxpool}_3(u) - \text{minpool}_3(u)$. This is an explicit boundary term; no Tversky recall-bias claim is made. Adaptive positive weighting, when audited, safely sets weight one for a zero-positive batch. The sampling ablation uses replacement probability $q_i = w_i / \sum_j w_j$, foreground ratio r_i , Canny edge ratio c_i , and $s_i = 1.5$ if $r_i < 0.05$ (otherwise 1):

$$w_i = \begin{cases} 0.3, & r_i \leq 0.001, \\ (2 + 3r_i)s_i(1 + c_i), & r_i > 0.001. \end{cases} \quad (2)$$



Training: $0.5\mathcal{L}_{Dice} + 0.5\mathcal{L}_{BCE}$; boundary control adds $0.1\mathcal{L}_{Dice}(B(p), B(y))$. Controls remove/replace orange modules and SCSE.

Fig. 1. Implemented 2-D semantic segmentation path. Normalized intensity is replicated to three channels and augmented by the maximum response of a 24-filter Gabor bank. A learned projection preserves compatibility with the pretrained encoder; UNet++/SCSE produces a probability map restored to the native image grid. Matched controls remove or replace the orange components and SCSE.

2.2. Data and leakage controls

BBBC039v1 contains 200 16-bit 520×696 U2OS Hoechst fields (about 23,000 nuclei) from one chemical screen [5, 6]. We use the provider’s 100/50/50 train/validation/test split. No crops or ROIs are generated. Identifiers, source groups and decoded-image SHA-256 hashes are checked across partitions. The validation set has 49 foreground fields and one empty field; the test set has no empty references. BBBC039 partly overlaps BBBC038, so BBBC038 is not claimed as independent validation.

TNBC v1.1 provides 50 H&E images and binary nuclear masks from 11 patients [7]. All images are used exactly once as a post-hoc zero-shot stress test; patient identifiers define inference units. RGB entry and explicit OpenCV grayscale produce bit-identical tensors for all 50 TNBC and 65 BBBC038 images because the pipeline already converts every image to grayscale. Since BBBC039 nuclei are bright on dark backgrounds while H&E nuclei are dark on bright tissue, we additionally evaluate one fixed transformation, $x' = 255 - x$, with no fitting, threshold tuning or checkpoint selection. TNBC labels must not be reused to select further preprocessing.

The private UWM lineage contains 855 valid annotated 2-D planes exported from in-house MCF10A confocal z -stacks (Zeiss LSM, $40 \times /1.1$ W, 405/461 nm, $0.2506 \mu\text{m}$ pixels, $1.0 \mu\text{m}$ z -step, 16-bit). One trained annotator delineated foreground with expert review; no repeat-annotation agreement study is available. Exact provenance links 334 planes to eight named stacks; the remaining 521 lack retained acquisition identifiers. We therefore report its corrected analysis as an audit, using 23 conservative geometry-defined groups split 516/133/206 planes (11/5/7 groups), not as independent external validation. Corrected trained models use three seeds and matched 30-epoch budgets. Every valid mask contains foreground; the earlier 60% empty statement arose from thresholding 0/1 masks at 127. No private images are distributed or required for the public-study claims.

2.3. Training and statistical analysis

Every main configuration uses seeds 0/1/2, 20 epochs, 512^2 inputs, batch size 8, AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), cosine decay, gradient clipping at 1, and the same flips and 90° rotations. The highest complete native-grid validation Dice selects a checkpoint; the final short batch is retained. Threshold 0.5 and no test-time augmentation are fixed for every model. Thus the seven component models (21 runs), ten validation-only sensitivities (30 runs) and six architecture baselines (18 runs) total 69 runs/1,380 epochs.

Predictions are restored to the native grid before Dice, IoU, boundary F1 (BF1) and surface Dice at two pixels, and bidirectional HD95. Both-empty pairs score Dice/IoU/BF1 one and HD95 zero; one-empty pairs score overlap/boundary zero and image-diagonal HD95. Foreground and empty strata are reported separately. For paired comparisons, seed scores are averaged within each field, 10,000 paired bootstrap resamples form 95% CIs, and two-sided Wilcoxon tests receive Holm correction. TNBC is first averaged within patient. Seed SD quantifies training variation, not biological sample size.

3. RESULTS

The Gabor bundle gains 0.250 Dice points over vanilla B7 (95% CI 0.197–0.299; Holm $p = 7.34 \times 10^{-9}$), but has only a -0.014 point difference from the strongest matched baseline, UNet++/Xception/SCSE (CI -0.047 – 0.018 ; Holm $p = .378$). The boundary term and residual fusion do not improve the bundle. Cellpose-SAM obtains 96.95% semantic Dice as a frozen reference, but unknown pretraining overlap and unequal training make it a contextual comparator rather than a fair baseline.

The validation-only sensitivity study never decodes test cases. Its 8-orientation/3-frequency reference obtains 96.684% foreground Dice. Four and 12 orientations yield 96.686% and 96.664%; one and five frequencies yield

(a) Private UWM result reconciliation		
Analysis / model	Dice (95% CI)	IoU
Rejected-paper stored aggregate [†]	.9513	.9025
Mean of its 128 saved scores [†]	.9413	.8925
Corrected grouped full pipeline	.9312 [.9134,.9472]	.8739
Corrected vanilla B7/U-Net++	.9349 [.9175,.9515]	.8805
Corrected plain Dice+BCE	.9366 [.9199,.9529]	.8831
Corrected U-Net/ResNet-50	.9320 [.9140,.9502]	.8752
Cellpose-SAM, frozen reference	.9198 [.9058,.9321]	.8539

(b) Published public-dataset context	
Study / protocol	Reported endpoint
Ours, BBBC039 official split	semantic Dice .9685
Caicedo U-Net, official split [8]	object F1 .898
SegmentGraph, official split [9]	Dice .9482; AJI .8680
ASW-Net, expanded interiors [10]	DICE1 .9645
MA-Net, BBBC039 [11]	Dice .973
NUSeg, fivefold CV [12]	Dice .9693; IoU .9406
Ours, all-50 TNBC zero-shot+invert	semantic Dice .5130
CellTranspose, TNBC 3–10 shot [13]	pixel-F1 .670–.757
Cellpose, selected transfer [13]	pixel-F1 .5829

Table 1. Private-study reconciliation and public literature context. (a) The rejected-paper aggregate is one percentage point above the mean recomputed from its saved arrays; both used a leakage-prone slice split. Corrected rows use 206 test planes in seven held-out geometry groups, group-macro metrics, three seeds for trained models and group-bootstrap CIs; none of the full-pipeline comparisons is significant after Holm correction. (b) Metrics, splits, selection, postprocessing and adaptation differ, so these published values are not a rankable leaderboard.

Table 2. Matched BBBC039 test results (three-seed mean; Dice includes seed SD). All use the same split and budget.

Model	Params(M)	Dice(%)	IoU(%)	BF1(%)
Vanilla B7	68.2	96.60±.07	93.44	98.53
Gabor+projection+SCSE	68.4	96.85±.13	93.90	98.63
Gabor+boundary	68.4	96.84±.05	93.88	98.68
SCSE-only B7	68.4	96.38±.53	93.06	98.48
Residual Gabor B7	68.4	96.57±.27	93.40	98.52
Residual Sobel B7	68.4	96.44±.48	93.17	98.47
Residual Gabor+boundary	68.4	96.61±.31	93.48	98.58
U-Net/ResNet-34	24.4	96.73±.29	93.68	98.71
U-Net/ResNet-50	32.5	96.74±.21	93.69	98.67
UNet++/EfficientNet-B0	6.6	96.59±.06	93.40	98.55
UNet++/Xception/SCSE	35.3	96.87±.11	93.93	98.69
DeepLabV3+/ResNet-50	26.7	95.05±.10	90.57	96.93
FPN/ResNet-50	26.1	94.97±.28	90.43	96.96

96.723% and 96.715%; $\sigma = 3$ and 7 yield 96.541% and 96.709%; Sobel, neutral-edge and weighted-sampling controls yield 96.727%, 96.586% and 96.731%. Weighted sampling improves foreground Dice by 0.047 points (CI 0.022–0.080) but performs worse on the single empty image, so it is not adopted. These small effects do not support unique optimality of the 24-filter bank.

On TNBC, explicit grayscale exactly reproduces prior results: vanilla B7 is best among the seven configurations at $8.07 \pm 1.39\%$ patient-macro Dice. Fixed inversion raises vanilla B7 to $51.30 \pm 0.40\%$, a 43.24-point patient-paired gain (95% CI 36.55–50.15; Holm $p = .00684$). The Gabor bundle rises from 2.94% to 37.66%. Residual over-segmentation confirms that correcting polarity does not solve stain, morphology or tissue-context shift.

4. DISCUSSION AND CONCLUSION

The corrected evidence supports a narrow conclusion: hand-crafted edges can yield a small within-acquisition gain over one vanilla B7 control, but architecture choice can match that

gain and domain shift overwhelms it. Inversion is a useful diagnostic because it isolates contrast polarity with no learned parameters; its large TNBC effect also exposes the danger of declaring generalization from fluorescence alone.

This study resolves the prior review’s computational concerns through public splits, explicit empty handling, equal budgets, SOTA architecture controls, sensitivity ablations, confidence intervals, boundary metrics, complete failures and reproducible code. It also makes the private lineage auditable rather than repeating its optimistic headline: the rejected paper’s 95.13% aggregate does not equal the 94.13% mean of its saved scores, and under corrected grouping the proposed additions do not beat vanilla or plain-loss controls (Table 1a). It does not resolve biological validation. BBBC039 is one U2OS screen, TNBC is post-hoc and label-examined, UWM source identifiers are incomplete, three seeds are limited, and binary masks do not support touching-instance AP. A decisive follow-up should lock polarity normalization on training/validation data and evaluate once on an unseen patient-separated cohort. Existing 3-D results remain preliminary and are not evidence for this 2-D claim.

5. COMPLIANCE WITH ETHICAL STANDARDS

BBBC039 is an openly released cell-line benchmark. TNBC is previously published, de-identified public histology reused without intervention, interaction, access to identifiers or attempted re-identification. No private patient data are analyzed. **The authors must obtain a formal UWM determination and replace this sentence with its exact category, wording and identifier before submission.**

6. ACKNOWLEDGMENTS AND DISCLOSURES

This work was supported by U.S. Department of Energy grant DE-SC0025403. The authors declare no competing interests. OpenAI Codex assisted with repository auditing, analysis scripts, statistical summaries, figure preparation and language editing. The authors selected the research questions, ran and verified the experiments, inspected all outputs and take responsibility for the manuscript. Figures contain measured data and real images/predictions, not generated microscopy content. Code, manifests and per-image results are publicly available at <https://github.com/vkd14/breast-seg-isbi2027>.

7. REFERENCES

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *MICCAI*, 2015, pp. 234–241.
- [2] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang, “UNet++: A nested U-Net architecture for medical image segmentation,” in *DLMI*, 2018, pp. 3–11.
- [3] Marius Pachitariu, Mike Rariden, and Carsen Stringer, “Cellpose-SAM: Superhuman generalization for cellular segmentation,” *bioRxiv*, 2025.
- [4] Anwai Archit et al., “Segment anything for microscopy,” *Nature Methods*, vol. 22, pp. 579–591, 2025.
- [5] Vebjorn Ljosa, Katherine L. Sokolnicki, and Anne E. Carpenter, “Annotated high-throughput microscopy image sets for validation,” *Nature Methods*, vol. 9, no. 7, pp. 637, 2012.
- [6] Juan C. Caicedo et al., “Nucleus segmentation across imaging experiments: The 2018 data science bowl,” *Nature Methods*, vol. 16, pp. 1247–1253, 2019.
- [7] Peter Naylor, Marick Lae, Fabien Rey, and Thomas Walter, “Segmentation of nuclei in histopathology images by deep regression of the distance map,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 2, pp. 448–459, 2019.
- [8] Juan C. Caicedo et al., “Evaluation of deep learning strategies for nucleus segmentation in fluorescence images,” *Cytometry Part A*, vol. 95, no. 9, pp. 952–965, 2019.
- [9] Jingru Yi, Pengxiang Wu, Menglin Jiang, Qiaoying Huang, Daniel J. Hoepfner, and Dimitris N. Metaxas, “SegmentGraph: A new framework for nuclei segmentation,” in *IJCAI*, 2022, pp. 1348–1354.
- [10] Hui Wang et al., “ASW-Net: An asymmetric sampling wide network for nuclei segmentation in fluorescence microscopy images,” *BMC Bioinformatics*, vol. 23, pp. 117, 2022.
- [11] Sultan Alotaibi et al., “A multi-scale attention network for nuclei segmentation,” *PLOS ONE*, vol. 19, no. 8, pp. e0308326, 2024.
- [12] Chathurika S. Wickramasinghe et al., “NUSeg: A deep learning framework for nuclei segmentation in microscopy images,” *Advanced Intelligent Systems*, 2024.
- [13] Yufan He et al., “CellTranspose: Few-shot domain adaptation for cellular segmentation,” in *WACV*, 2023, pp. 455–466.